PEFL: A Privacy-Enhanced Federated Learning Framework for Mobile Edge CrowdSensing in the Presence of Collusion and Backdoor Attacks

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***Abstract*—Mobile Edge CrowdSensing (MECS) based on Federated Learning (FL) has attracted widespread attention as an intelligent data collection and processing approach. FL trains the global model through aggregating local models of participants without requiring the exchange of raw data. However, directly sharing local models is vulnerable to backdoor attacks launched by adversaries. What's worse, malicious server may collude with participants to manipulate the parameter updating of models and even compromise the accuracy of whole system. To address these challenges, this paper proposes a Privacy-Enhanced Federated Learning (PEFL) framework for MECS with the aim of resisting both backdoor and collusion attacks. In PEFL, a Backdoor Resistant Privacy-Enhanced Aggregation (BRPEA) mechanism with the Differential Privacy-Enhanced Exponential (DPEE) method is developed to perturb local models of participants. Clustering and clipping techniques are also designed in BRPEA to effectively distinguish backdoor models from the benign local models, which eliminate the influence of local models deviation and optimize the noise introduced by differential privacy. Furthermore, a Collusion Resistant Privacy-Preserving Aggregation (CRPEA) mechanism is studied. CRPEA can avoid the collusion between servers and participants and prevent the privacy of local models from being leaked. The theoretical analysis proves the security of proposed PEFL framework and the simulation experiments demonstrate that PEFL can not only ensure the aggregation accuracy of encrypted models but provide robustness against both backdoor and collusion attacks.**

***Index Terms*—Mobile CrowdSensing, federated learning, backdoor attacks, collusion attacks, differential privacy.**

I. Introduction

With the rapid evolution of wireless communication technologies, Mobile CrowdSensing (MCS) [1] has emerged as a transformative approach to data collection which utilizes various mobile devices (e.g., smartphones, in-vehicle computers and wearable electronics) and heterogeneous sensors (e.g., visual cameras, microphones and speedometers) to collect, process and analyze data [2]. Empowered by the framework of MCS, Mobile Edge CrowdSensing (MECS) integrates its distributed computing capabilities of edge nodes to establish a fundamental architecture for distributed model training. The advent of advanced mobile communication networks and accelerated development of 'Internet of Everything' [3] have created new demands for greater data processing accuracy, reduced errors and optimized service performance.

MECS has shown great potential in areas such as intelligent transportation [4] and emergency response [5]. Its distributed architecture effectively improves data processing efficiency and system responsiveness. In practical applications, MECS usually assumes that Edge Servers (ESs) and participants are semi-honest entities. They follow system protocols and honestly perform sensing tasks but may be curious about other information [6]. In addition, the large number of heterogeneous devices in MECS provides opportunities for attackers to disguise themselves as benign participants to compromise. However, existing defense mechanisms are ineffective when ESs cooperate with multiple malicious participants or these participants share information to jointly manipulate model behavior. They also detect hidden backdoors implanted that maintain global model performance during the local training phase. Therefore, it is urgent to propose a robust defense mechanism against the collusion and backdoor attacks to comprehensively improve the security and credibility in MECS.

MECS is more vulnerable to collusion attack under Federated Learning (FL) since ESs and participants cannot be fully trusted. Existing privacy preserving techniques, such as Secure Multi-party Computation (SMC) [7], Differential Privacy (DP) [8],

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Homomorphic Encryption (HE) [9], [10] and Digital Signature Mechanism (DSM) [11], provide certain protections for MECS. However, they typically assume that ESs are fully honest. This overlook the possibility that dishonest ESs may collude with malicious participants, which implies the urgent need for resilient mechanisms to resist collusion attacks.

Backdoor attack [12], [13] are also a major challenge in MECS systems due to its invisibility. Researchers have developed defense to enhance the protection against backdoor attack including backdoor filtering methods [14], [15], differential privacy perturbation techniques [16], [17], robust aggregation algorithms [18], [19], knowledge distillation [20] and anomaly update detection approaches [21], [22]. However, some existing studies hold that ESs are fully trustworthy and overlook potential collusion between the ESs and malicious participants. Therefore, most mechanisms limit the defense effectiveness in MECS and it is necessary to build a defense mechanism that simultaneously mitigates backdoor and collusion attacks.

Although existing defense mechanisms have improved data security in MECS, some key challenges remain to be solved. (i) DP adds random noise to local model parameters to protect the data privacy of participants, which may compromise the robustness of global model; (ii) Two-Trapdoor Homomorphic Encryption (TTHE) reduce the impact of collusion between malicious ESs and participants, but it cannot fully prevent accuracy degradation; (iii) Most existing solutions focus on addressing either collusion or backdoor attacks and often rely on strong trust assumptions regarding ESs. We aims to design a comprehensive framework that mitigates both threats while balancing privacy preservation and model accuracy.

In this paper, a Privacy-Enhanced Federated Learning (PEFL) framework is proposed to handle the aforementioned challenges in MECS. Firstly, we constructed a Backdoor Resistant Privacy-Enhanced Aggregation (BRPEA) mechanism. This mechanism perturbs local model parameters by designing a Differential Privacy-Enhanced Exponential (DPEE) method to mitigate the impact of backdoor attack on the security of global model. In addition, model clustering and clipping techniques were proposed to further detect potential malicious models so as to ensure the robustness of MECS. Secondly, we developed a Collusion Resistant Privacy-Preserving Aggregation (CRPEA) mechanism to defend against the collusion attack. Different from existing studies, we integrate the secret sharing and Two-Trapdoor Homomorphic Encryption (TTHE) to guarantee the confidentiality of model parameters in model aggregation for FL. The proposed PEFL not only enhances the accuracy of global model but also significantly improves the data privacy of FL in MECS. The main contributions can be summarized as follows.

- We propose a novel PEFL framework for MECS that simultaneously defends against both backdoor and collusion attacks. Unlike existing methods that address only a single threat or assume trusted ESs, PEFL ensures both local model accuracy and participant privacy in adversarial settings.
- BRPEA mechanism applies DPEE to perturb the parameters of local model in order to prevent the leakage of sensitive data. In addition, it employs model clustering to identify backdoor local models and further utilizes clipping technique to limit the amplification of local model weights. Consequently, it effectively resists the security threat of malicious participants to global model.
- CRPEA mechanism is developed to cope with collusion attack during local model updating in MECS. This mechanism combines TTHE with blinding factor technique to encrypt local models. It effectively prevents ESs and malicious participants from inferring private data of benign participants and ensures the privacy and accuracy of local models during the data transmission and aggregation.

The structure of this paper is organized as follows: Section II reviews the related work, while Section III outlines the network and threat models of the proposed framework. Section IV discusses the knowledge of techniques used in this paper, and Section V presents the framework and implementation of PEFL. Section VI provides the security analysis, followed by experimental validation in Section VII. Finally, Section VIII concludes this paper and summarizes the proposed framework.

II. Related Work

Although FL has been widely applied in MECS to protect the data privacy during the data sensing and processing in a distributed way, it remains vulnerable to various security threats, especially collusion [23] and backdoor [24] attacks.

A. Collusion Defenses

Faced with the challenging issues of collusion attack in MECS, various methods have been explored, such as SMC, DSM and HE. Sun et al. proposed bilateral privacy protection scheme based on adaptive location generalization and group aggregation to reduce the risk of collusion between malicious participants and external attackers [25]. Yan et al. introduced privacy-preserving collaborative machine learning scheme that aimed at preventing information leakage and defending against malicious participants. The scheme employed random permutations in Software Guard Extensions (SGE) to prevent local model parameters from being accessed by the curious server and the dishonest participants [26]. Xiong et al. presented a FL crowdsensing scheme using smart contract, HE and pseudonymization defend against collusion attack to enhance data privacy [27]. Gao et al. proposed lightweight commitment scheme with irreversible gradient transformations to protect the data privacy of participants. They also suggested verification mechanism based on optimized Lagrange interpolation to detect whether the server was involved in collusion attacks [28]. Chen et al. focused on the security of accessing data in vehicular networks and a server-assisted attribute signature mechanism was proposed to resist collusion attack between participants and servers [29]. Compared with existing privacy-preserving schemes that only address a single type of attack, PEFL innovatively considers the combined threat of collusion and backdoor attacks. It enhances privacy-preserving and ensures the reliability and accuracy of global model.

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TABLE I
COMPARISON WITH EXISTING SCHEMES

Scheme	Attack Type	Privacy Preserving Technique	Assumed Condition
Copif [27]	Collusion attack	Homomorphic encryption and pseudo-identity	Untrusted servers and participants
Vcd-FL [28]	Collusion attack	Lightweight commitment	Malicious servers
ROPE [30]	Backdoor attack	PCA, expectation maximization and isolation forest	Untrusted participants
FLAME [34]	Backdoor attack	Model clustering and weight clipping	Untrusted participants
ShieldFL [35]	Model poisoning	Two-trapdoor homomorphic encryption	Malicious participants
P2BroFL [36]	Model Poisoning	Local model selection with privacy and top-k protocol	Malicious servers and participants
ProFL [37]	Byzantine attack	Linearly homomorphic encryption and secure protocols	Byzantine participants
PEFL (Ours)	Backdoor and collusion attacks	Differential privacy and two-trapdoor homomorphic encryption with improved blinding factor	Curious-but-honest servers and participants

B. Backdoor Defenses

Backdoor attack can be categorized into data poisoning attack and model tampering attack according to different attack targets. In the studies regarding data poisoning attack, Principal Component Analysis (PCA) and similarity-based approaches have been proposed in the past few years. Wang et al. used PCA to extract critical gradient features and applied expectation maximization to detect malicious participants. They also developed an isolation forest method to identify and remove malicious gradients from backdoor attack [30]. Hou et al. proposed similarity-based method to protect the integrity of deep learning models. This method detected anomalies and defends backdoor attack based on the similarity between suspicious samples and predefined validation samples [31]. Concerning the model tampering attack, defense mechanisms including stability, knowledge distillation and robust aggregation are proposed as feasible solutions. Xue et al. proposed a backdoor defense method which incorporated with soft knowledge distillation to further enhance the purification of backdoor in Deep Neural Networks (DNNs) [32]. Gao et al. designed a FL scheme to defend backdoor attack by introducing a secure partial aggregation protocol. This protocol restricted participants from directly uploading local models to server, ensuring that the server cannot infer private information from the encrypted local models [33]. Nguyen et al. proposed a defense framework named FLAME to remove backdoor through injecting adequate noise [34].

In contrast to the aforementioned literature, which focuses on a single type of attack or relies on strict assumptions, PEFL can effectively defend against backdoor and collusion attacks in MECS while achieving both privacy preserving and model accuracy. To clearly demonstrate the differences between existing mechanisms and PEFL, we compare them from three dimensions: attack types, privacy preserving techniques and assumed conditions, as shown in Table I.

TABLE II
COMMONLY USED SYMBOLS

Symbol	Definition
G^t	Global model in round t
ω_i^t	Local model of each participant in round t
$\bar{\omega}_i^t$	Local model perturbed by DPEE in round t
$\bar{\omega}_i'^t$	Local model by clipping in round t
D_i	Local dataset of each participant
ℓ, p, q	Parameters for the cryptosystem
(K_p, λ), (K_p^*, λ^*)	Key pairs generated by TP
λ_1 and λ_2	Two parts of the private key $K_p = \lambda$
$[\![m]\!]$	Ciphertext from plaintext m
Φ_{jk}	Shared key
H	Secure one-way hash function
R_j	Blinding factor for P_j
r_j	Random number for P_j
$\cos_{ab}$	Cosine similarity between local models $\bar{\omega}_a^t$ and $\bar{\omega}_b^t$
e_i	Magnitude difference
S^t	Clipping bound
K	Total number of participants
N	Number of participants selected by FL
L	Number of benign participants

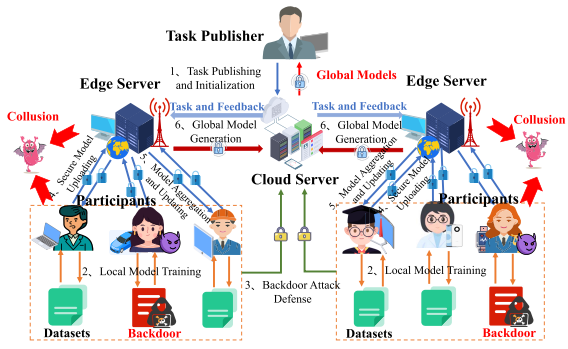


Fig. 1. Task execution framework of PEFL in MECS.

III. NETWORK AND THREAT MODEL

This section outlines the fundamental concepts of PEFL and the symbols with their definitions used in this paper are listed in Table II for easy understanding.

A. Network Model

We present the detailed PEFL framework based on FL for MECS in Fig. 1. PEFL simultaneously outsources both data sensing and processing tasks to participants, which enables full utilization of their computation and storage resources. It comprises four entities: Task Publisher (TP), Cloud Server (CS), Edge Servers (ESs) and participants.

- *Task Publisher (TP):* In MECS, TP generates two key pairs, (K_p, λ) and (K_p^*, λ^*), for the secure model uploading and final result generation stages, respectively. TP is responsible for assigning data sensing tasks and distributing the partial private keys to participants.

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- *Cloud Server (CS):* TP distributes sensing tasks to CS, which divides task into multiple subtasks and assigns them to the corresponding ESs.
- *Edge Servers (ESs):* ESs take charge of recruiting participants to initiate FL training. In the t-th round, each ES selects N participants from the participant set $P = \{p_i | 1 \leq i \leq K\})$ to form a subset $P_N = \{p_j | 1 \leq j \leq N\}$ where $P_N \subseteq P$ and $N \leq K$. ESs aggregate local models of these selected participants p_j to generate global model which is then shared with these participants for the training of next round. ESs will send global models to CS when the termination condition is met.
- *Participants:* A participant p_i executes data collection and sensing tasks via intelligent devices. After receiving the subtask issued by ESs, participants encrypt the local models using the blinding factors combined with TTHE and submit them to ESs.

B. Threaten Model

Although some mechanisms against different attacks have been introduced in previous literature about FL-based MECS, they focus on a single attack pattern and little work has been devoted to the scenario that malicious ESs may collude with participants to mislead local models even the global model of FL in MECS. In our system model, TP is fully trusted, whereas ESs and participants are semi-honest. They follow the system protocol but may attempt to infer private information to manipulate FL process. Faced with the new challenge in MECS along with the development of FL, collusion and backdoor attacks are explored in the threaten model in order to design a more comprehensive countermeasure.

1) Collusion Attack: Collusion attack is a type of internal attack that malicious participants may collude with ESs to cause more harmful impact on MECS than attacks carried out by the malicious participant or ESs independently. In this paper, we focus on the scenario where ESs collude with fewer than $(L-1)$ malicious participants to access the private data of benign participants. Specifically, ESs hold part of the private key λ_2, while participant p_i holds part of the private key λ_1. During the global model aggregation, benign participant p_b uploads the encrypted local model $[\![\omega_b^t]\!]$ to ESs. ESs then use their λ_2 to collude with λ_1 held by the malicious participant $p_{mal(mal \neq b)}$ to decrypt $[\![\omega_b^t]\!]$, obtain local model of p_b and manipulate the parameter of p_b. This compromises private data and undermines the security of FL model in MECS.

2) Backdoor Attack: An attacker may control one or more participants which are either malicious participants or benign participants during the training of local model. The attacker can manipulate the parameters of local models, decrease the model accuracy and disclose the data privacy of MECS. Assumed that the attacker can influence less than $N/2$ participants where N is the number of participants selected for FL and damage the training data or parameters of these participants, it cannot control the operations on ESs or interfere with the benign participants.

IV. Privacy Preserving Theory in PEFL

This section briefly describes the fundmental theory of PEFL used in this paper, including two-trapdoor homomorphic encryption system, blinding factor technique and differential privacy mechanism.

A. Two-Trapdoor Homomorphic Encryption System

TTHE system based on the Paillier encryption is designed to handle the issue of authorization in multi-key environments [35]. It comprises six algorithms as follows.

- *Key Generation (KeyGen) Algorithm:* $(K_p, K_s) \leftarrow KeyGen(1^\ell)$ Let p and q be two large prime numbers, each with a bit size $\mathbb{L}(p) = \mathbb{L}(q) = \ell$. The key generation algorithm computes $n = p \cdot q$ and $\lambda = lcm(p-1, q-1)$ using the security parameter ℓ as the input where $lcm(\cdot)$ represents the least common multiple. The modular inverse function is defined as $u = (L((1+\lambda \cdot n) mod n^2))^{-1}$, where the function $L(u) = \frac{u-1}{n}$. The public key is $K_p = n$ and the corresponding private key is $K_s = \lambda$.
- *Key Splitting (KeySplit) Algorithm:* $(\lambda_1, \lambda_2) \leftarrow KeySplit(K_s = \lambda)$ The key splitting algorithm randomly divides the private key $K_s = \lambda$ into two parts, λ_1 and λ_2 that meet the requirements of $\lambda_1 + \lambda_2 \equiv 0 \pmod{\lambda}$ and $\lambda_1 + \lambda_2 \equiv 1 \pmod{n}$.
- *Encryption (Enc) Algorithm:* $[\![m]\!] \leftarrow Enc(m, K_p)$ Given a plaintext $m \in \mathbb{Z}_n$ and a public key K_p, the encryption algorithm selects a random number $r \in \mathbb{Z}_n^*$ and generates the ciphertext $[\![m]\!] \leftarrow Enc(m, K_p) = (1 + m \cdot n) \cdot r^n \mod n^2$.
- *Decryption (Dec) Algorithm:* $m_{dec} \leftarrow Dec([\![m]\!], K_s = \lambda)$ If a entity receives a ciphertext $[\![m]\!]$ and holds the private key K_s, it recovers the plaintext $m_{dec} \leftarrow L([\![m]\!]^\lambda \mod n^2) \cdot u \mod n$ using its private key K_s.
- *Partial Decryption (PartDec) Algorithm:* $\hat{[\![m]\!]}_i \leftarrow PartDec([\![m]\!], \lambda_i)$ Given the ciphertext $[\![m]\!]$ and the partial private key $\lambda_i (i = 1, 2)$ generated by KeySplit algorithm, the corresponding ciphertext for partial decryption is $\hat{[\![m]\!]}_i \leftarrow [\![m]\!]^{\lambda_i} \mod n^2$.
- *Plaintext Extraction (FullDec) Algorithm:* $m_{full} \leftarrow FullDec(\hat{[\![m]\!]}_1, \hat{[\![m]\!]}_2)$ Assumed $\hat{[\![m]\!]}_1$ and $\hat{[\![m]\!]}_2$ are two partial decrypted ciphertexts generated by PartDec algorithm, the plaintext m_{full} is extracted using $m_{full} \leftarrow L(\hat{[\![m]\!]}_1 \cdot \hat{[\![m]\!]}_2 \mod n^2)$ where $L(u) = \frac{u-1}{n}$.

Noticed that the plaintext m_{full} obtained by decrypting the ciphertext using $PartDec$ and $FullDec$ algorithms is equivalent to the plaintext m_{dec} obtained by decrypting the ciphertext using Dec algorithm. Let m_1 and m_2 be two plaintexts and $[\![m_1]\!]$ and $[\![m_2]\!]$ be the ciphertexts encrypted using the public key K_p. Any two ciphertexts encrypted with the same public key satisfy the additive and multiplicative homomorphic properties which can be represented as (1) and (2), respectively.

$$[\![m_1]\!] \cdot [\![m_2]\!] = [\![m_1 + m_2]\!] \tag{1}$$

$$[\![m_1 \cdot m_2]\!] = [\![m_1]\!]^{m_2} \tag{2}$$

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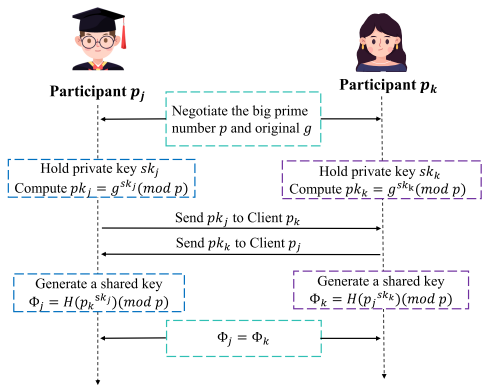


Fig. 2. Hashed Diffie-Hellman key exchange protocol.

B. Blinding Factor Technique

We consider that the sensitive data may be leaked if TTHE system is directly applied because ESs hold the same partial private key and may collude with participants.To address this issue, we introduce the Hashed Diffie-Hellman Key Exchange (HKE) protocol [38] as shown in Fig. 2.

Based on the HKE protocol, participant $p_j (j \in \{1, \ldots, N\})$ generates a shared key Φ_{jk} using (3).

$$\Phi_{jk} = H\left(pk_k^{sk_j}\right) \mod n, k \in 1, \ldots, N and k \neq j \quad (3)$$

where shared key Φ_{jk} is obtained according to a hash function $H(\cdot)$. HKE protocol can ensure that a shared key can be negotiated between participants through an insecure communication channel and the third-party eavesdropper cannot know the shared key. Subsequently, an improved blinding factor R_j can be derived based on the shared key using (4).

$$\begin{aligned} R_j &= \sum_{k=1,k\neq j}^{N} (-1)^{\beta} \cdot \Phi_{jk} \mod n \\ &= \sum_{k=1,k\neq j}^{N} (-1)^{\beta} \cdot H\left(pk_k^{sk_j}\right) \mod n \end{aligned} \quad (4)$$

where $\beta = (j + k)$ if $k < j$ and $\beta = (j + k - 1)$ if $k > j$.

The aggregated ciphertext still satisfies the homomorphic property according to the additive and multiplicative homomorphism in (1) and (2). Taking four participants as an instance, the participants compute four blinding factors using (4), R_1, R_2, R_3 and R_4. These blinding factors satisfy $\sum_{j=1}^{4} R_j = (R_1 + R_2 + R_3 + R_4) \mod n = 0$, so $\prod_{j=1}^{4} R_j = 1 \mod n^2$. Namely, it follows (5).

$$\begin{aligned} &[\![m_1]\!] \odot [\![m_2]\!] \odot [\![m_3]\!] \odot [\![m_4]\!] \\ &= \left(1 + n \cdot \sum_{j=1}^{4} m_j\right) \left(\prod_{j=1}^{4} r_j\right)^n \cdot \prod_{j=1}^{4} R_j \mod n^2 \\ &= \left(1 + n \cdot \sum_{i=1}^{4} m_i\right) \cdot r^n \mod n^2 \\ &= [\![m_1 + m_2 + m_3 + m_4]\!] \end{aligned} \quad (5)$$

where $\odot$ stand for aggregation operation. Eq. (5) proves that the aggregated ciphertext is still a valid homomorphic ciphertext.

The blinding factor R_j introduced in TTHE system is a random mask designed to improve randomness during the encryption process. This ensures that ESs cannot decrypt alone or collude with malicious participants to infer the original data of benign participants even if they holds part of decryption key. We improve the traditional TTHE system discussed in Section IV-A and develop a collusion resistant privacy enhancing aggregation mechanism.

C. Privacy Clustering and Clipping Optimization

In this subsection, we introduce the Differential Privacy Exponential Mechanism, Model Clustering and Adaptive Clipping to protect the privacy of participants and improve the stability of the global model.

1) Differential Privacy: Definition 1 (ϵ-DP [39]): A privacy preserving mechanism M satisfies ϵ-DP if for any datasets D and D', and any subset δ_i of Ψ, we have,

$$Pr[M(D) \in \delta_i] \leq exp(\epsilon) \times Pr[M(D') \in \delta_i] \quad (6)$$

where $\epsilon > 0$ is the privacy budget, Ψ represents all possible outputs of M. D and D' represent two similar datasets that only one data point is different.

Definition 2 (Exponential Mechanism [40]): Given a scoring function $q : (D, \delta_i)$ for a dataset D, which evaluates the quality of output $\delta_i \in \Psi$, if the exponential mechanism M satisfies ϵ-DP, then

$$M(D, q) = \left\{\delta_i | P(\delta_i \in \Psi) \propto \exp\left(\frac{\epsilon q(D, \delta_i)}{2\Delta q}\right)\right\} \quad (7)$$

where Δq is the global sensitivity of scoring function $q(D, \delta_i)$.

The exponential mechanism does not directly return a deterministic result but randomly outputs a value according to a probability distribution. Eq. (8) is used to normalized calculate the probability of each candidate result.

$$P(\delta_i \in \Psi) = \frac{\exp\left(\frac{\epsilon q(D,\delta_i)}{2\Delta q}\right)}{\sum_i \exp\left(\frac{\epsilon q(D,\delta_i)}{2\Delta q}\right)} \quad (8)$$

Gaussian mechanism achieves (ϵ, δ)-DP by adding noise $N(0, \sigma^2)$ to continuous model parameters. This mechanism has been adopted in FLAME [34] to eliminate the influence of backdoor attacks. However, it relies on the parameter δ and there is still a potential risk of being exploited by attackers. Different from the Gaussian mechanism, exponential mechanism provides ϵ-DP by sampling from discrete candidate sets. The sampling probability for each candidate is determined by a scoring function together with the privacy budget ϵ. The exponential mechanism not only ensures the privacy but also enhances the resistance to backdoor attacks in the scenario of discrete optimization.

2) Model Clustering: The model clustering classifies local models by separating them into different clusters, thereby distinguishing backdoor models from benign ones. HDBSCAN employs a hierarchical clustering method to form a stable cluster structure across varying density levels. It groups closely related

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data points in n-dimensional space into clusters. Data points not assigned to any cluster are marked as outliers, which are subsequently identified as backdoor local models. As HDBSCAN does not require predefined parameters, it is more suitable for the complex environments of MECS.

However, a limitation of our method is that it may not detect all backdoor models. An attacker can control the angle and amplitude deviation of model updates, making the malicious models appear highly similar to benign ones. HDBSCAN is designed primarily to eliminate high-impact malicious models with large angular deviations in the update vector space, rather than to detect all malicious models.

The cosine similarity is introduced to measure angular differences among all local models and serves as the input feature of model clustering in this paper. By analyzing clustering results, the backdoor local models and the benign local models can be distinguished according to angle deviations. In the set of participants P, the cosine similarity between the local models $\bar{\omega}_b^t$ and $\bar{\omega}_b^t$ of two participants p_a and p_b is defined as (9).

$$\cos_{ab} = \frac{\bar{\omega}_a^t \cdot \bar{\omega}_b^t}{\|\bar{\omega}_a^t\| \|\bar{\omega}_b^t\|} \tag{9}$$

where $\bar{\omega}_a^t$ and $\bar{\omega}_b^t$ represents the perturbed results of the local models ω_a^t and ω_b^t after applying DPEE. $\bar{\omega}_a^t \cdot \bar{\omega}_b^t$ is the dot product of perturbed local models, and $\|\bar{\omega}_a^t\| \|\bar{\omega}_b^t\|$ is the product of their Euclidean distances to form the cosine similarity $\cos_{ab}$. $\cos_{ab}$ can be converted into a distance matrix $c_{ab} = 1 - \cos_{ab}$ and applied to HDBSCAN algorithm to optimize the clustering process for local models. If a local model is not assigned to any cluster, it denotes that the local model is significantly different from other benign models within a cluster and it will be labeled as a backdoor local model.

3) Model Clipping: Although the model clustering method can detect the backdoor local models with large angular deviations, the model clipping is necessary technique to amend the weights of local models. The selection of clipping boundary, S^t, is a critical issue. If S^t is too large, attackers may greatly impact on the performance of global model in terms of convergence through expanding the weights of local models; if S^t is too small, some benign local models will be over-clipped, which decreases the accuracy of global model. Therefore, a reasonable model clipping technique should be proposed and the clipping boundary will be dynamically adjusted according to the distribution of local models.

The manhattan distance is a common method to measure the distance between two local model vectors. For participant p_i, the amplitude difference between its local model $\bar{\omega}_i^t$ of current training round t and its local model $\bar{\omega}_i^{t-1}$ of previous training round can be determined using (10).

$$e_i = \|\bar{\omega}_i^t - \bar{\omega}_i^{t-1}\|_1 = \sum_{y=1}^{Y} |\bar{\omega}_{i,y}^t - \bar{\omega}_{i,y}^{t-1}| \tag{10}$$

where $\|\cdot\|_1$ represents the L1 norm. Y is the dimension of local model vector and represents the number of parameters of each local model.

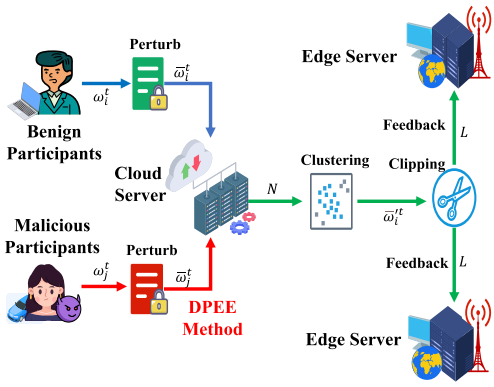


Fig. 3. Backdoor attack defense framework of PEFL in MECS.

The proposed mechanism and methods in this section not only protect the data privacy and defend against attacks in MECS, but also maximize the contribution of benign models.

V. Privacy-Enhanced Aggregation Scheme

This section will comprehensively discuss our proposed PEFL framework to resist both the backdoor and collusion attacks in MECS. The motivation of PEFL is to protect the privacy of local model parameters submitted by participants without losing the accuracy of global model. PEFL consists of two key mechanisms, the Backdoor Resistant Privacy-Enhanced Aggregation (BRPEA) mechanism and the Collusion Resistant Privacy-Preserving Aggregation (CRPEA) mechanism to enhance security of MECS system.

A. Backdoor Resistant Privacy-Enhanced Aggregation Mechanism

During federated training in MECS, participants may suffer from backdoor attack with magnitude and angle deviations. We develop a BRPEA mechanism in PEFL on the basis of DP mentioned in Section IV-C to distinguish the backdoor models from benign one. It ensures that the local models uploaded to ESs cannot contain backdoor models. Fig. 3 illustrates the workflow of BRPEA mechanism in MECS.

1) BRPEA Mechanism - DPEE: The backdoor attack usually has a certain degree of concealment and even a slight magnitude or an angular deviation may result in great impact on the accuracy of global model. The exponential mechanism described in Section IV-C1 is adopted to perturb the local models. The randomness of perturbation introduces the variability into the model selection, which strengthens the privacy of local models and reduces the threat of backdoor attacks.

Any participant p_i receives the global model from ESs in each round t of FL and trains this model based on its local sensing data D_i. Each training round will involve $\Gamma (1 \leq \Gamma \leq \lfloor D_i/B \rfloor)$ local iterations to obtain a weighted local model ω_i^t. The participant applies DPEE method to perturb the local model parameters $\omega_i^\chi(\kappa)$ and constructs a perturbed local model $\bar{\omega}_i^t$. The detailed process can be formalized as follows.

In the t-th training round, the sampling boundary serves as the critical factor in determining the sampling range which reflects

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the changing direction and amplitude of the current and previous iteration parameters. For each parameter $\omega_i^{\chi}(\kappa)$ in the local model ω_i^t, the sampling boundary can be obtained according to (11).

$$X = \omega_i^{\chi}(\kappa) - \omega_i^{\chi-1}(\kappa) \tag{11}$$

where $\omega_i^{\chi-1}(\kappa)$ is the κ-th local model parameter in the $\chi-1$ iterations of round t. The sampling boundary X decides whether the sampling range F should be adjusted upward or downward, which helps implement DP in the local model. The sampling range F is determined by (12).

$$F = \begin{cases} [\omega_i^{\chi}(\kappa) - |X|, \omega_i^{\chi}(\kappa)], & X < 0 \\ [\omega_i^{\chi}(\kappa), \omega_i^{\chi}(\kappa) + X], & \text{otherwise} \end{cases} \tag{12}$$

At each iteration χ, participant p_i calculates local model parameters for each instance κ in batch B and constructs a candidate parameter set $\mathbb{S}$ with Ω elements based on (12). This set can protect the privacy of model parameters without compromising their availability.

$$\mathbb{S} = \begin{cases} \left[\omega_i^{\chi}(\kappa) - |X|, \omega_i^{\chi}(\kappa) - \frac{(\Omega-1)|X|}{\Omega}, \ldots, \omega_i^{\chi}(\kappa) - \frac{|X|}{\Omega}\right], & X<0 \\ \left[\omega_i^{\chi}(\kappa) + \frac{X}{\Omega}, \omega_i^{\chi}(\kappa) + \frac{2X}{\Omega}, \ldots, \omega_i^{\chi}(\kappa) + X\right], & \text{otherwise} \end{cases} \tag{13}$$

BRPEA defines a scoring function to evaluate the resistance of local model parameters to backdoor attack in (14).

$$q(\bar{\omega}_i^{\chi}(\kappa), \omega_i^{\chi}(\kappa)) = \|\bar{\omega}_i^{\chi}(\kappa) - \omega_i^{\chi}(\kappa)\|_1 \tag{14}$$

where $\bar{\omega}_i^{\chi}(\kappa)$ and $\omega_i^{\chi}(\kappa)$ denote the perturbed and the trained local model parameters, respectively. Based on (13) and (14), the probability of $\omega_i^{\chi}(\kappa)$ being perturbed to $\bar{\omega}_i^{\chi}(\kappa) \in \mathbb{S}$ is obtained using the exponential mechanism in (8).

$$P(M(\omega_i^{\chi}(\kappa)) = \bar{\omega}_i^{\chi}(\kappa)) = \frac{\exp\left(\frac{-\eta\epsilon}{2} q(\cdot)\right)}{\sum_{\bar{\omega}_i^{\chi}(\kappa) \in \mathbb{S}} \exp\left(\frac{-\eta\epsilon}{2} q(\cdot)\right)} \tag{15}$$

where η represents the sensitivity adjustment factor and ϵ is the privacy budget. In the χ-th iteration, exponential mechanism is used to sample the perturbed parameter values $\bar{\omega}_i^{\chi}(\kappa)$ from $\mathbb{S}$ for each parameter based on the probability distribution defined by (15). This perturbation process is repeated for all parameters to construct the perturbed local model $\bar{\omega}_i^{\chi}$ for the current iteration.

The weighted aggregation method assigns different weights to different iterations of local model. The perturbed local model for t-th round can be formalized as

$$\bar{\omega}_i^t = \sum_{\chi=0}^{\Gamma-1} \varsigma_i^{\chi} \bar{\omega}_i^{\chi} \tag{16}$$

where ς_i^{χ} is the weight coefficient which satisfies $\sum_{\chi=0}^{\Gamma-1} \varsigma_i^{\chi} = 1$, $\bar{\omega}_i^t$ represents the weighted aggregation of all perturbed local models across Γ local iterations in round t.

2) BRPEA Mechanism - Clustering: The model clustering based on HDBSCAN discussed in Section IV-C2 is introduced to further enhance the ability of PEFL in MECS against backdoor attack, and it includes three key steps as follows.

- The distance constraint is essential to distinguish the backdoor local models from the benign ones before the clustering method (Algorithm 2 in Section V-C1) is applied. Therefore, we adopt the cosine similarity among perturbed local models and transform the similarity between any two perturbed local models into a distance matrix as the distance constraint for HDBSCAN algorithm. Let p_i and p_j be any two participants in P_N, the cosine similarity between perturbed local models $\bar{\omega}_a^t$ and $\bar{\omega}_b^t$ is measured by cos_{ab} in (9) and the distance matrix can be represented as

$$\mathbb{C} = [c_{ab}]_{N\times N},$$
$$c_{ab} = 1 - \cos_{ab} = 1 - \frac{\bar{\omega}_a^t \bar{\omega}_b^t}{\|\bar{\omega}_a^t\| \|\bar{\omega}_b^t\|}, a, b = 1, \ldots, N \tag{17}$$

  where $\cos_{ab} \in [-1, 1]$, and each element $c_{ab} \in [0, 2]$ in the distance matrix $\mathbb{C}$. If $c_{ab} = 0$, it means completely similar; If $c_{ab} = 2$, it implies completely opposite. The distance matrix $\mathbb{C}$ will be used as the input of HDBSCAN algorithm for clustering analysis.
- HDBSCN algorithm sets the minimum cluster size, denoted as $min_cluster_size$, as a key parameter to ensure that each cluster contains most of benign local models. HDBSCN prevents overly small clusters or noise points from being mistaken for valid clusters by setting $min_cluster_size$ to at least $N/2+1$. In addition, HDBSCN algorithm sets $min_{samples} = 1$ as the minimum number of neighborhood samples for core point determination, balancing the stability of clustering and the identification of noise points.
- HDBSCAN algorithm is employed to cluster perturbed local models using the distance matrix $\mathbb{C}$ generated by (17). The local models are divided into benign local models and backdoor local models, where the index set of benign local models is recorded as L according to the clustering results of HDBSCAN.

This clustering technique can effectively distinguish benign and potential backdoor local models. It provides an important basis for subsequent local model clipping.

3) BRPEA Mechanism - Clipping: The clipping technique discussed in Section IV-C3 avoids the overfiting of global model through limiting the excessive deviation of weights. The proposed approach in this section further improves both the security and the convergence accuracy of global model.

BRPEA mechanism of PEFL introduces the clipping technique for the scaling of weights for local models based on HDBSCAN clustering. It obtains the index set L of benign local models from perturbed local models and evaluates the manhattan distance e_i between the local model $\bar{\omega}_i^t$ and the model $\bar{\omega}_i^{t-1}$ from the previous training round using (10). The set of Manhattan distances $E = \{e_1, \ldots, e_L\}$ is constructed and sorted in ascending order and the median of E is selected as the clipping boundary $S^t = Median(E)$ in the t-th training round. BRPEA mechanism assesses the scaling factor for each benign local model $(\bar{\omega}_1^t, \ldots, \bar{\omega}_L^t)$ after the clipping boundary S^t is determined. Taking the L-th local model $\bar{\omega}_L^t$ as an instance,

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its scaling factor ξ_L is defined as

$$\xi_L = \frac{S^t}{e_L} \tag{18}$$

where e_L is the manhattan distance of the L-th local model. Then, we have the weights of the L-th local model,

$$\hat{\omega}_L^t = \bar{\omega}_L^t \times min(1, S^t / \|\bar{\omega}_L^t\|_1) \tag{19}$$

In (19), $\hat{\omega}_L^t = \bar{\omega}_L^t$ if $\|\bar{\omega}_L^t\|_1 < S^t$. This indicates that a benign local model is unchanged; otherwise, the benign local model is scaled to $S^t/\|\bar{\omega}_L^t\|_1$ times if $\|\bar{\omega}_L^t\|_1 > S^t$.

B. Collusion Resistant Privacy-Preserving Aggregation Mechanism

In this section, CRPEA mechanism in PEFL is designed to resist collusion attack during local model updating with the aim of improving the security of FL in MECS.

After completing the detection of all local models using the BRPEA mechanism, the cloud server instructs each ES to receive all the participants uploads by L verified benign models. This effectively prevents backdoor local models from being included in the global model aggregation process of FL on ESs. CRPEA mechanism adopts the blinding factor technique proposed in Section IV-B to defend against attacks in the condition that no more than $L-1$ participants collude with ESs. The mechanism integrates TTHE with the blinding factor to ensure that the local model information of benign participants will not be leaked to ESs in the process of local models aggregation even in the presence of collusion attacks.

Firstly, each participant p_i weights the local model based on its weight coefficient ς_i^χ to obtain the weighted local model after training the local model ω_i^χ.

$$\omega_i^t = \sum_{\chi=0}^{\Gamma-1} \varsigma_i^\chi \cdot \omega_i^\chi \tag{20}$$

where, ς_i^χ denotes weight coefficient for the χ-th local iteration, and ω_i^χ represents the local model of participant p_i. Local model ω_i^t is obtained by performing a weighted aggregation over the Γ local iterations in t-th training round.

Secondly, participants encrypt their local models using TTHE in combination with the blinding factor according to (4). Specifically, each participant P_j selects a random number $r_j \in \mathbb{Z}_n^*$ and encrypts local model ω_j^t into a ciphertext $[\![\omega_j^t]\!]$. Noticed that the blinding factor R_j is embedded within the ciphertext.

$$[\![\omega_j^t]\!] = (1 + \omega_j^t \cdot n) \cdot r_j^n \cdot R_j \mod n^2 \tag{21}$$

In the t-th taining of global model, L participants encrypt their local models $(\omega_1^t, \omega_2^t, \ldots, \omega_L^t)$ using the public key K_p and generate the ciphertexts $([\![\omega_1^t]\!], [\![\omega_2^t]\!], \ldots, [\![\omega_L^t]\!])$ which will be uploaded to each ES. Each ES aggregates L encrypted local models based on (22) to obtain the ciphertext $[\![G^t]\!]$ of global model after it receives the ciphertexts of local models.

$$[\![G^t]\!] = \frac{L}{N} \cdot \{[\![\omega_1^t]\!] \odot [\![\omega_2^t]\!] \odot \cdots \odot [\![\omega_L^t]\!]\} = \frac{L}{N} \cdot \sum_{j=1}^{L} [\![\omega_j^t]\!] \tag{22}$$

In the next iteration of participants, ESs apply the partial key λ_2 to execute the $PartDec$ algorithm and obtain the partial decryption result $[\![\hat{G}^t]\!]_1 \leftarrow PartDec([\![G^t]\!], \lambda_2)$. ESs broadcast $([\![G^t]\!], [\![\hat{G}^t]\!]_1)$ to all participants in MECS. Each participant performs the partial decryption $[\![\hat{G}^t]\!]_2 \leftarrow PartDec([\![G^t]\!], \lambda_1)$ depending on the partial key λ_1 after receiving the broadcasted ciphertext. Finally, the participant merges the two partial decryption, $[\![\hat{G}^t]\!]_1$ and $[\![\hat{G}^t]\!]_2$, using the $FullDec([\![\hat{G}^t]\!]_1, [\![\hat{G}^t]\!]_2)$ algorithm discussed in Section IV-A to construct the final aggregation G^t in t-th training round.

$$G^t = \frac{L}{N} \cdot \sum_{j=1}^{L} \omega_j^t \tag{23}$$

Obviously, ESs cannot decrypt a ciphertext $[\![\omega_j^t]\!]$ due to the embedding of blinding factor R_j in HKE protocol unless ESs collude with more than or equal to $L-1$ participants. CRPEA significantly enhances the security of MECS and effectively protects the privacy for FL in MECS.

C. Privacy-Enhanced Aggregation Scheme

We present the detailed construction of PEFL framework for MECS in this section. The proposed framework consists of six steps, the task publishing and initialization, the local model training, the backdoor attack defense, the secure model uploading, the model aggregation and updating, and the global model generation as illustrated in Fig. 1 of Section III-A.

1) Task Publishing and Initialization: TP firstly issues a data sensing and processing task. Then, it selects a security parameter ℓ and executes the $KeyGen$ algorithm described in Section IV-A to generate two key pairs (K_p, λ) and (K_p^*, λ^*). TP calls the $KeySplit$ algorithm to split the private key λ into two parts, λ_1 and λ_2. Furthermore, it employs the hash function $H : \{0,1\}^* \rightarrow \mathbb{Z}_n^*$ to implement HKE protocol. Finally, TP publishes the public key (K_p, K_p^*), distributes the partial private key λ_2 to ESs and assigns the partial private key λ_1 along with the hash function H to all participants.

2) Local Model Training: Each participant p_i collects the sensing data using the sensors in smart devices and forms a local dataset D_i. ESs initialize FL global model G^0 and distribute it to all participants. In the t-th training round, each participant p_i trains its local model on dataset D_i to determine the gradient $\nabla\iota(G^{t-1}; B)$ after receiving the global model G^{t-1} from ESs. The participant p_i updates the local model parameters $\omega_i^\chi(\kappa)$ according to (24).

$$\omega_i^\chi(\kappa) \leftarrow G^{t-1} - \eta \cdot \nabla\iota(G^{t-1}; B) \tag{24}$$

where $\nabla\iota(G^{t-1}; B)$ represents the gradient of global model G^{t-1} related to loss function ∇, B is the batch size of training data D_i. Additionally, η is the learning rate which regulates the updating step size and $\omega_i^\chi(\kappa)$ is the local model parameter in t-th training round, local iteration χ and batch κ.

3) Backdoor Attack Defense: The backdoor attack defense contains three steps, DPEE model perturbing, HDBSCAN clustering and model clipping. Firstly, the participant $p_i (i \in \{1, \ldots, N\})$ applies DPEE method to perturb the local model

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Algorithm 1: DPEE Method.

Input: Local data D_i, Batch size B, Local model ω_i^t, Sensitivity factor η.
Output: Perturbed model $\bar{\omega}_i^t$ for p_i.

Participants calculate the local training iteration $\Gamma \leftarrow |D_i|/B$;
for $t \in \{1, 2, \ldots, T\}$ **do**
  for $\chi \in \{0, 1, \ldots, \Gamma - 1\}$ **do**
    for $\kappa \in \{1, \ldots, B\}$ **do**
      Participant p_i computes sampling boundary $X = \omega_i^{\chi}(\kappa) - \omega_i^{\chi-1}(\kappa)$;
      p_i uses Eq.(12) to obtain sampling range F;
      Generate candidate set $\mathbb{S}$ by Eq.(13);
      for $i \in \mathbb{S}$ **do**
        Executing Eq.(15) obtains $\bar{\omega}_i^{\chi}(\kappa) \leftarrow Perturbation(\omega_i^{\chi}(\kappa), \epsilon)$;
      end
    end
  end
end
Compute final perturbed local model $\bar{\omega}_i^t = \sum_{\chi=0}^{\Gamma-1} \zeta_{\chi} \bar{\omega}_i^{\chi}$;
return $\bar{\omega}_i^t$.

ω_i^t and the designed scoring function $q(\bar{\omega}_i^{\chi}(\kappa), \omega_i^{\chi}(\kappa))$ based on (14) to reinforce the resistance of global model when backdoor attack emerges. Secondly, the participant p_i computes the probability that $\omega_i^{\chi}(\kappa)$ is perturbed to $\bar{\omega}_i^{\chi}(\kappa)$ based on the scoring function and the probability defined in (15). Eventually, all perturbed local models in every iteration χ are assigned the different weights and the final perturbed local model $\bar{\omega}_i^t$ of p_i is constructed. Every participant p_i executes Algorithm 1 and all perturbed local models are sent to cloud sever for model clustering and clipping.

HDBSCAN method described in Algorithm 2 clusters all the perturbed local models to reduce the impact of backdoor attack. The cosine similarity $\cos_{ab}$ between the perturbed local models is evaluated based on (9). The similarities of all participants are transformed into a distance matrix $\mathbb{C}$ as shown in (17) which is applied into HDBSCAN clustering to distinguish benign local models from backdoor local models.

After perturbed local models are clustered, the clipping technique dynamically adjusts the scaling factor to limit the negative impact of malicious local model on the convergence of global model. Assumed L be an index set of benign local models identified by Algorithm 2, cloud server firstly assesses the manhattan distance e_l (where $l \in I$) between $\bar{\omega}_l^t$ and $\bar{\omega}_l^{t-1}$ of P_l using (10). Cloud server stores the manhattan distance in set E and sorts them in an ascending order. It selects the median value as the clipping threshold S^t in the t-th training round. Based on e_l and S^t, cloud server further determines the scaling factor ξ_l for each local model. It applies clipping technique demonstrated in Algorithm 3 to clip benign local models, which enhances robustness and stability of global model aggregation.

Algorithm 2: HDBSCAN Clustering.

Input: Perturbed local models $\bar{\omega}_i^t (i \in \{1, \ldots, N\})$.
Output: Benign local models $\bar{\omega}_i^t (i \in \{1, \ldots, L\})$ after clustering.

Calculate cosine similarity $\cos_{ab} \leftarrow CosineSimilarity(\bar{\omega}_1^t, \ldots, \bar{\omega}_N^t)$ using (9);
Compute distance matrix $\mathbb{C} = [c_{ab}]_{N \times N}$ by (17), where $c_{ab} = 1 - \cos_{ab}$;
Set clustering parameters at least $min_cluster_size = \lceil N/2 \rceil + 1$ and $min_{samples} = 1$;
Obtain labels via HDBSCAN clustering $Labels = HDBSCAN(\mathbb{C}, min_{samples}, min_cluster_size)$;
Extract benign models $L = count(Labels \neq -1)$;
return L benign local models $\bar{\omega}_i^t (i \in \{1, \ldots, L\})$.

Algorithm 3: Clipping Technique.

Input: L perturbed local models $\bar{\omega}_i^t$.
Output: Clipped benign local models $\bar{\omega}_l^{'t}$ held by TP.

Set an index set $L = (1, \ldots, L)$;
for $l \in [1, L]$ **do**
  Compute manhattan distance $e_l = \|\bar{\omega}_l^t - \bar{\omega}_l^{t-1}\|_1 = \sum_{y=1}^{Y} |\bar{\omega}_{l,y}^t - \bar{\omega}_{l,y}^{t-1}|$;
  Add e_l to set E;
end
Sort the set $E = Sort\{e_1, \ldots, e_L, L\}$;
Select the median as the clipping boundary $S^t = Median(E)$;
for $l \in [1, L]$ **do**
  Determine the scaling factor $\xi_l = \frac{S^t}{e_l}$;
  if $\|\bar{\omega}_l^t\|_1 < S^t$ **then**
    $\bar{\omega}_l^{'t} = \bar{\omega}_l^t$;
  end
  $\hat{\omega}_L^t = \bar{\omega}_l^t \times min(1, \frac{S^t}{e_l})$;
end
return Benign local models $\bar{\omega}_l^{'t}$.

4) Secure Model Uploading: ESs select L participants for FL training based on the clipped benign local models according to BRPEA mechanism. Firstly, $p_{j,(j \in 1,\ldots,L)}$ takes the average of all batch parameters and weights the local model based on its coefficient ς_i^{χ} to obtain the local model $\omega_i^t = \sum_{\chi=0}^{\Gamma-1} \varsigma_i^{\chi} \cdot \omega_i^{\chi}$. Secondly, each participant p_j randomly selects a confidential value $s_j \in \mathbb{Z}_{\rho}^*$ as the private key sk_j and publishes the public key $pk_j = g^{s_j} \mod \rho$ for p_j according to HKE protocol. The participant p_j computes Φ_{jk} using (3) and constructs the blinding factor R_j to further enhance the security of interaction between ESs and participants.

Algorithm 4 illustrates the encryption process of local model. Finally, the participant p_j uses the blinding factor R_j and the public key K_p broadcasted by TP to encrypt local model ω_j^t against collusion attack between ESs and participants. Noticed that the local model ω_j^t is encrypted into $[\![\omega_j^t]\!]$ using (21) in Line

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Algorithm 4: Model Encryption.

Input: Local model $\omega_j^t (j \in \{1, \ldots, L\})$.
Output: Encryption model $[\![\omega_j^t]\!]$ for p_j.

$R_j \leftarrow 0$;
for $k > 1$ *to* L, $k \neq j$ **do**
  if $k < j$ **then**
    $\beta = j + k$;
  end
  else if $k > j$ **then**
    $\beta = j + k - 1$;
  end
  $R_j = R_j + (-1)^{\beta} \cdot H\left(pk_k^{sk_j}\right) \mod n$;
end
return $[\![\omega_j^t]\!] = (1 + \omega_j^t \cdot n) \cdot r_j^n \cdot R_j \mod n^2$.

Algorithm 5: Global Model Updating.

Input: Encryption local model $[\![\omega_j^t]\!]$, participants with (K_p, λ_1) and ESs with (K_p, λ_2).
Output: Participants obtain global model G^t.

Each ES aggregates local models $[\![G^t]\!] = \frac{L}{N} \cdot \sum_{i=1}^{L} [\![\omega_j^t]\!]$;
Each ES calls $PartDec$ to partially decrypt global model $[\![\hat{G}^t]\!]_1 \leftarrow [\![G^t]\!]^{\lambda_2} \mod n^2$ using λ_2;
Each ES sends $([\![G^t]\!], [\![\hat{G}^t]\!]_1)$ to all participants;
Participants call $PartDec$ to get partial global model $[\![\hat{G}^t]\!]_2 \leftarrow [\![G^t]\!]^{\lambda_1} \mod n^2$ using λ_1 and $FullDec$ to decrypt global model $G_t = L([\![\hat{G}^t]\!]_1 \cdot [\![\hat{G}^t]\!]_2 \mod n^2)$;
return G^t to participants.

11 in order to protect the private parameters of local models, which we have discussed in Section V-B. During the t-th training round, each participant p_j encrypts the local model and uploads the ciphertext $[\![\omega_j^t]\!]$ to ESs instead of directly sending the local model.

5) Model Aggregation and Updating: The model aggregation and updating includes two parts, local model aggregation and global model updating. Initially, Each ES aggregates the ciphertext of local model $[\![\omega_j^t]\!]$ which is uploaded by participant p_j to generate the ciphertext of global model $[\![G^t]\!]$ based on (22). In next training round of FL, ES broadcasts the ciphertext of global model and the ciphertext of partial decryption, $[\![G^t]\!]$ and $[\![\hat{G}^t]\!]_1$, to all participants.

Then, each ES uses the partial private key λ_2 to perform $PartDec$ algorithm. However, ESs can only obtain the partial decrypted ciphertext of global model, denoted as $[\![\hat{G}^t]\!]_1 \leftarrow PartDec([\![G^t]\!], \lambda_2)$.

Next, each participant p_i receives the ciphertext of global model and the partial decrypted ciphertext $([\![G^t]\!], [\![\hat{G}^t]\!]_1)$. Participant p_i executes $PartDec$ algorithm to generate another partial decrypted ciphertext $[\![\hat{G}^t]\!]_2 \leftarrow PartDec([\![G^t]\!], \lambda_1)$ using its partial private key λ_1. Finally, the $FullDec$ algorithm combines two partial decrypted ciphertexts $[\![\hat{G}^t]\!]_1$ and $[\![\hat{G}^t]\!]_2$ to recover the global model G^t. This can be expressed as $G^t \leftarrow FullDec([\![\hat{G}^t]\!]_1, [\![\hat{G}^t]\!]_2)$.

In fact, both ESs and participants regard the updated global model G^t as the initial model for the $(t+1)$-th training round. The iteration will continue until the predefined termination condition T is satisfied. The detailed procedure for the updating of global model is outlined in Algorithm 5.

6) Global Model Generation: The framework iteratively performs from local model training in Section V-C2 to global model aggregation in Section V-C5 until the maximum number of iterations T is reached. The participant p_j encrypts the local model ω_j^T into $[\![\omega_j^T]\!]^*$ which will be transmitted to ESs using the public key K_p^* from the key pair (K_p^*, λ^*). Each ES aggregates the received the ciphertexts of local models to generate the ciphertexts of global model $[\![G^T]\!]^* = [\![\omega_1^t]\!]^* \odot [\![\omega_2^t]\!]^* \odot \cdots \odot [\![\omega_L^t]\!]^*$ and send them to cloud sever. The cloud server forwards the global model $[\![G^T]\!]^*$ received from each edge server to TP. Finally, TP decrypts the global model ciphertext $[\![G^T]\!]^*$ using the private key λ^* from the key pair (K_p^*, λ^*). As a result, TP obtains the final global model G^T as shown in (25).

$$G^T = \frac{L}{N} \cdot \sum_{j=1}^{L} \omega_i^T \tag{25}$$

After verifying the privacy and security of global model G^T, TP analyzes the data in MECS and distributes them to other heterogeneous platforms through security protocols to support cross domain collaborative inference and service deployment.

VI. Theoretical Analysis

In this section, we prove the security of CRPEA and BRPEA mechanisms in PEFL and analyze the computational and communication complexity in PEFL.

A. Security Analysis

1) Security of CRPEA: Definition 3 (Hash Diffie-Hellman (HDH) Assumption). Let $\mathbb{G} = \langle g \rangle$ be a multiplicative cyclic group with the prime order p and the generator g. $H : \{0,1\}^* \rightarrow \{0,1\}^{\ell}$ is a hash function where ℓ is a security parameter. Given two quadruples $T_1 = (g, g^a, g^b, H(g^{ab}))$ and $T_1 = (g, g^a, g^b, H(g^z))$ for $a, b, z \xleftarrow{R} \mathbb{Z}_p$. It is computationally infeasible for any Probabilistic Polynomial-Time (PPT) adversary A to distinguish T_1 from T_2, that is,

$$Adv_A^{HDH} = |Pr[A(T_1) = 1] - Pr[A(T_2) = 1]| \leq \epsilon(\ell) \tag{26}$$

where $\epsilon(\ell)$ is a negligible function on security parameter ℓ [41].

Based on **Definition 3**, we firstly prove the semantic security of CRPEA mechanism in MECS.

Theorem 1: CRPEA mechanism in Section V-B is semantically secure if **Defination 3** holds and ESs collude with fewer than or equal to $L - 2$ participants.

Proof 1: Within an MECS system, K denotes the total number of participants and $N (N \leq K)$ participants are selected by FL. Among these N participants, L $(L \leq N)$ participants are

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verified as benign ones who qualify for the encrypted model aggregation. Therefore, the security boundary of CRPEA depends on these L benign participants.

When ESs collude with fewer than or equal to $L-2$ participants, we assume that any two honest participants, p_x and p_y cannot take part in the collusion attack and each participant p_j $(1 \leq j \leq L)$ has its plaintext of ω_i. Obviously, ESs can obtain the private key sk_c through colluding with $L-2$ colluders ($P_c = \{p_j | j \neq x, y\}$) to decrypt the aggregated ciphertext $[\![\omega_x]\!] \odot [\![\omega_y]\!]$ uploaded by p_x and p_y according to Dec algorithm of TTHE in Section IV-A. ESs only obtain the aggregation ciphertext but can not distinguish the local model of any participant, which satisfies the computational inseparability. We further prove that the local model of P_x is secure in CRPEA when ESs collude with fewer than or equal to $L-2$ participants. Based on (4) and (21), the plaintext ω_x is calculated as (27).

$$M_x = (1 + \omega_x \cdot n) \cdot r^n \cdot R_x \tag{27}$$

The colluders ($P_c = \{p_j | j \neq x, y\}$) can infer the blinding factor R_x of participant P_x through sharing their private key sk_j and random number r_j. Therefore, the colluders can infer the ciphertext of P_x based on (28),

$$M'_x = (1 + (\omega_x + r_t \cdot H(g^{s_x} g^{s_y})) \cdot n) \cdot r^n \tag{28}$$

where r_t is a random number related to the t-th training round, $H(g^{s_x} g^{s_y})$ is a hash value of HDH and r^n is a random number. ESs can obtain the private key sk_c through colluding with $L-2$ colluders. However, they can only decrypt the cipertext M'_x and recover the plaintext $\omega_x + r_t \cdot H(g^{s_x} g^{s_y})$ rather than the plaintext ω_x.

It should be emphasized that the above theorem establishes a security boundary. When the number of colluders is nearly $L-2$, MECS system enters a state of single point of failure and its security relies entirely on the two remaining benign participants. If the random number or private key of either party is leaked, the local model could be recovered, making the system extremely vulnerable. Although computational indistinguishability holds in theory, the probability of successful recovery of local models by colluders increases significantly and the overall security of the system decreases sharply when the proportion of colluders $(L-2)/L$ approaches 1. In summary, CRPEA mechanism can ensure that the local models of participants are not disclosed to ESs as long as the number of colluders is less than $L-2$.

Theorem 1 indicates the security of CRPEA and we will further prove the performance of CRPEA in privacy preserving to ensure that the plaintext ω_x of participant p_x is secure against any PPT adversary A.

Lemma 1: If the Decisional Composite Residuosity (DCR) assumption holds, then TTHE is semantically secure [42].

Definition 4 (Simulation-Based Security): For any PPT adversary A, if there is a simulator C^* for all possible inputs W, the view $REAL$ of adversary A in the real world is computationally indistinguishable from the view $IDEAL$ of simulator C^* in the ideal world, we can assert that the mechanism Π in (29) is secure [43].

$$REAL_A^{\Pi}(W) \equiv_c IDEAL_{C^*}^{\Pi}(W') \tag{29}$$

where $\equiv_c$ denotes the indistinguishability in computation, W and W' are sets of parameters for the real world and the ideal world, respectively.

Theorem 2: In PEFL framework, CRPEA mechanism is capable of securely aggregating the encrypted local models even in the presence of PPT simulator C^*, which makes ESs cannot obtain any information about the local models trained by participants and their sensing data.

Proof 2: CRPEA mechanism constructs an indistinguishable security proof between the real world and the ideal world based on the semantic security of TTHE system in Lemma 1. This provides the security of cryptosystem which defends against the PPT adversaries to distinguish the real world from the ideal world with a non-negligible advantage.

Firstly, Definition 4 assumes that there is a challenger C and an adversary A and they interact with each other in the real world $REAL_A^{\Pi}$. In the task publishing and initialization phase of CRPEA mechanism, C generates the system parameters $(K_p, \lambda_1, \lambda_2)$ and sends the leakage function $\mathcal{L} = \{K_p, g, \rho, \lambda_1, \lambda_2, sk_j, \omega_j^t, G^t, H\}_{j \in \{1,\ldots,L\} \setminus \{x,y\}}$ to A. C encrypts the local models $\{\omega_1^t, \omega_2^t, \ldots, \omega_L^t\}$ of participants to the ciphertexts $\{[\![\omega_1^t]\!], [\![\omega_2^t]\!], \ldots, [\![\omega_L^t]\!]\}$ and transmits these ciphertexts to the adversary A. A aggregates all ciphertexts of local models and obtains the encrypted global model, $[\![G^t]\!] = \frac{L}{N} \cdot [\![\prod_{j=1}^{L} \omega_j^t]\!]$. Using the $PartDec$ and $FullDec$ algorithms in Section IV-A and the private keys λ_1 and λ_2, the adversary A decrypts $[\![G^t]\!]$ to recover the global model $G^t = \frac{L}{N} \cdot \sum_{j=1}^{L} \omega_j^t$. Therefore, A infers the global model G^t based on Algorithm 5.

Secondly, Definition 4 supposes that there is a simulator C^* which interacts with the adversary A in the ideal world $IDEAL_{C^*}^{\Pi}$. C^* provides the leakage function $\mathcal{L} = (K_p, g, \rho, \lambda_1, \lambda_2, sk_{j,j \in \{1,\ldots,L\} \setminus \{x,y\}}, H)$ to A and selects a random number r_w to simulate the local model of participant p_x. Then, the local model of participant p_y can be expressed as (30).

$$\omega_y^{t'} = \sum_{j=1}^{L} \omega_j^{t'} - \sum_{j \in P_c}^{L} \omega_j^{t'} - \omega_x^{t'} \tag{30}$$

where $\sum_{j=1}^{L} \omega_j^{t'}$ is the ideal global model and $\sum_{j \in P_c}^{L} \omega_j^{t'}$ is an aggregation model of colluders. C^* computes the ciphertexts, $[\![\omega_x^{t'}]\!]$ and $[\![\omega_y^{t'}]\!]$, using Enc algorithm. It can deduce the aggregation local models $[\![\sum_{j \in P_c}^{L} \omega_j^{t'}]\!]$ of the colluders $P_c = \{p_j | j \neq x, y\}$ and sends these encrypted local models to A. The adversary A can obtain the encrypted aggregation model $[\![G^{t'}]\!]$ of L participants as shown in (31).

$$\begin{aligned} [\![G^{t'}]\!] &= \frac{L}{N} \cdot [\![\omega_x^{t'}]\!] \odot [\![\omega_y^{t'}]\!] \odot \left[\!\!\left[\sum_{j \in P_c}^{L} \omega_j^{t'} \right]\!\!\right] \\ &= \frac{L}{N} \cdot \left[\!\!\left[\omega_x^{t'} + \omega_y^{t'} + \sum_{j \in P_c}^{L} \omega_j^{t'} \right]\!\!\right] \\ &= \frac{L}{N} \cdot \left[\!\!\left[\sum_{j=1}^{L} \omega_j^{t'} \right]\!\!\right] \end{aligned} \tag{31}$$

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Therefore, A recovers the global model $G^{t'} = \frac{L}{N} \cdot \sum_{j=1}^{L} \omega_j^{t'}$ based on Algorithm 5.

Finally, A cannot distinguish the real view $REAL_A^{\Pi}$ from the simulated view $IDEAL_{C^*}^{\Pi}$ because the global models, G^t and $G^{t'}$, recovered by the adversary A are equal. Therefore, these global models satisfy (32).

$$REAL_A^{\Pi}(\omega_x^t, \omega_y^t) \equiv_c IDEAL_{C^*}^{\Pi}(\omega_x^{t'}, \omega_y^{t'}) \tag{32}$$

That is, CRPEA mechanism is secure and it can resist collusion attacks according to Lemma 1 and Definition 4.

2) Security of BRPEA: CRPEA mechanism is developed based on BRPEA mechanism and the security of BRPEA should also be analyzed. Before discussing its security, we introduce two lemmas which will be used to prove the privacy performance of DPEE.

Lemma 2: Sequential Composition. Let a set of privacy mechanisms $\mathbb{M} = \{M_1, M_2, \ldots, M_n\}$ be sequentially performed on a dataset. $\mathbb{M}$ satisfies $\sum_i \epsilon_i - DP$ if each M_i satisfies $\epsilon_i - DP$ [44].

Lemma 3: Postprocessing. Supposed a mechanism $M : D \to R$ satisfies $\epsilon - DP$ and a function $f : R \to R'$ is an arbitrary randomized mapping. Then, $f \circ M : D \to R'$ still satisfies $\epsilon_i - DP$ [44].

Based on these lemmas, we will prove DPEE satisfies $(T \cdot \epsilon) - DP$ in Theorem 3.

Theorem 3: The proposed DPEE method satisfies $(T \cdot \epsilon) - DP$ where T is the number of iterations in local model training and ϵ is the privacy budget.

Proof 3: DPEE method applies the exponential mechanism of DP in Definition 2 during each training round t, where $0 \leq t \leq T$. Based on the theory of exponential mechanism, each parameter of local model satisfies $\epsilon - DP$ in t-th training round. According to Lemmas 2 and 3, each parameter of local model undergos T times of DP operations during T training rounds. Therefore, each local model parameter satisfies $(T \cdot \epsilon) - DP$.

In the next step, we discuss that BRPEA mechanism can protect the privacy of local models after **Theorem 3** is proved.

Theorem 4: In PEFL framework, BRPEA mechanism can defend against backdoor attack if the PPT adversary A emerges.

Proof 4: Firstly, Definition 4 indicates that C^* can perfectly simulate the parameters of perturbed local model $\bar{\omega}_i^{\chi'}(\kappa)$ using the random number $r \in \mathbb{Z}_n^*$ in an ideal world. Therefore, the real view $REAL_A^{\Pi}(\bar{\omega}_i^{\chi}(\kappa))$ and the simulated view $IDEAL_{C^*}^{\Pi}(\bar{\omega}_i^{\chi'}(\kappa))$ are indistinguishable for adversary A.

Secondly, DPEE perturbation method satifies $(T \cdot \epsilon) - DP$ and the invariance of postprocessing for DP, which ensures that any postprocessing applied to the perturbed local model $\bar{\omega}_i^{\chi'}(\kappa)$ does not compromise the privacy of this local model according to Theorem 3. Consequently, adversary A cannot distinguish the perturbed local models $\bar{\omega}_a^{\chi}$ in the real world from the corresponding model $\bar{\omega}_a^{\chi'}$ in the ideal world during any training round t for participant p_a. Similarly, adversary A is also incapable of differentiating $\bar{\omega}_b^{\chi}(\kappa)$ from $\bar{\omega}_b^{\chi'}(\kappa)$ for participant p_b. Furthermore, it is impossible for adversary A to distinguish the real view $REAL_A^{\Pi}(\mathbb{C})$ from the simulated view $IDEAL_{C^*}^{\Pi}(\mathbb{C}')$ where $\mathbb{C}$ and $\mathbb{C}'$ represent the distance matrices obtained in the real and ideal worlds.

Thirdly, the perturbed local model $\bar{\omega}_l^{t'}$ in the ideal world generated by the simulator C^* is also indistinguishable from the perturbed local model $\bar{\omega}_l^t$ in the real world for participant p_l. Accordingly, the manhattan distance e_l' determined by simulator C^* is identical to the manhattan distance e_l in the real-world, which guarantees that the scale factors ξ_l' and ξ_l are also consistent. In the ideal world, the simulator C^* amplifies or reduces the perturbed local model $\hat{\omega}_l^{t'} = \bar{\omega}_l^{t'} \times \min(1, S^{t'}/\xi_l')$, where $S^{t'}$ and ξ_l' denote the scaling factor and the clipping boundary obtained from the ideal world, respectively. The perturbation parameters, the distance matrix and the scaling operation remain consistent in the real and ideal worlds so that the clipped local model $\hat{\omega}_l^t$ is also computationally indistinguishable. Therefore, the following equation is satisfied.

$$\begin{aligned} &REAL_A^{\Pi}(\bar{\omega}_i^{\chi}(\kappa), \bar{\omega}_l^t, \mathbb{C}, e_l, S^t, \xi_l, \hat{\omega}_l^t) \equiv_c \\ &IDEAL_{C^*}^{\Pi}(\bar{\omega}_i^{\chi'}(\kappa), \bar{\omega}_l^{t'}, \mathbb{C}', e_l', S^{t'}, \xi_l', \hat{\omega}_l^{t'}) \end{aligned} \tag{33}$$

Finally, we conclude that BRPEA mechanism in PEFL can effectively defend against backdoor attacks because adversary A is unable to distinguish the real view $REAL_A^{\Pi}$ from the simulated view $IDEAL_{C^*}^{\Pi}$.

Based on the security proofs of CRPEA and BRPEA mechanism in this section, PEFL achieves the privacy preserving in MECS. In addition, CRPEA should be incorporated with BRPEA mechanism to guarantee the overall security of PEFL against both the backdoor and collusion attacks.

Lemma 4: A protocol is perfectly simulatable if all subprotocols of this protocol are perfectly simulatable [45].

Theorem 5: PEFL can preserve the privacy of local model for each participant p_i in MECS and realize the overall security of PEFL if the security attributes in both Theorems 2 and 4 are satisfied.

Proof 5: The local models are encrypted using CRPEA mechanism based on TTHE. Because ESs only possess a weak decryption key, it is infeasible to decrypt the ciphertexts of local models by ESs if BRPEA mechanism is provably secure. In addition, CRPEA mechanism is constructed based on BRPEA mechanism and its security under the collusion scenarios inherently relies on the security of BRPEA. Lemma 1 explains the security of TTHE and we have proven the security of CRPEA and BRPEA mechanisms in Theorems 2 and 4, respectively. The evidence, combined with the conclusion of Lemma 4, suggests that our proposed PEFL framework can protect the privacy of local models and ensure the overall security of MECS in a complex scenario with the backdoor and collusion attacks.

The security of our scheme is proven based on the assumption of a trusted third party (TP), as discussed in [6], [35] and [37]. However, a single TP introduces the risk of single point of failure and the issue of centralized trust. A scheme with multiple task publishers or a distributed key management mechanism based on threshold cryptography should be explored to further enhance the security of MECS.

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TABLE III
COMPUTATION AND COMMUNICATION COMPLEXITY ON PEFL

Phase	PEFL	
	Computation	Communication
Local Model Training	$\mathcal{O}(\mu \cdot \Gamma)$	$\mathcal{O}(0)$
Backdoor Attack Defense	$\mathcal{O}(N \cdot \mu) + \mathcal{O}(N \cdot \log N)$	$\mathcal{O}(N \cdot \mu)$
Secure Model Uploading and Aggregation	$\mathcal{O}(L \cdot \mu \cdot (T_{Mul} + T_{Add}) + \mu \cdot T_{PDec})$	$\mathcal{O}(\mu \cdot \lvert\mathbb{X}\rvert)$
Global Model Generation	$\mathcal{O}(\mu \cdot T_{Dec})$	$\mathcal{O}(2\mu \cdot \lvert\mathbb{X}\rvert)$

TABLE IV
COMPARISON OF DIFFERENT MECHANISMS UNTIL CONVERGENCE

Mechanism	Communication Round	Accuracy (%)	Time (min)
ShieldFL [35]	83	82.47	377.38
ProFL [37]	96	83.07	234.78
PEFL (Ours)	80	92.86	298.66

B. Complexity Analysis

In this part, we analyze the computational and communication complexity of each training round in PEFL, as summarized in Table III. Let N be the number of participants, L be the number of benign participants, Γ be the number of local training iterations, and μ be the dimension of the local model. Let $|\mathbb{X}|$ be the communication complexity of an encrypted number. T_{Enc}, T_{Dec}, T_{PDec}, T_{Mul} and T_{Add} denote the computational cost of encryption, decryption, partial decryption, homomorphic multiplication and addition, respectively.

During the local training phase, each participant independently performs Γ rounds of local model updates and it costs $\mathcal{O}(\mu \cdot \Gamma)$ for computation. In the backdoor attack defense phase, the ESs need to perform DPEE perturbations on N local models and complete clustering and clipping operations. The computational complexity is $\mathcal{O}(N \cdot \mu) + \mathcal{O}(N \cdot \log N)$ and the communication complexity is $\mathcal{O}(N \cdot \mu)$. Each ES needs to select L benign participants to perform m homomorphic additions and one partial decryption. The computational complexity is $\mathcal{O}(L \cdot \mu \cdot (T_{Mul} + T_{Add}) + \mu \cdot T_{PDec})$ and the communication complexity is $\mathcal{O}(\mu \cdot |\mathbb{X}|)$. Each ES sends the encrypted global model to cloud server, which forwards it to TP for decryption. The computational complexity is $\mathcal{O}(\mu \cdot T_{Dec})$ and the communication complexity is $\mathcal{O}(2\mu \cdot |\mathbb{X}|)$.

Based on the complexity analysis, we further compare the computational costs of ShieldFL, ProFL and PEFL in Table IV. The number of participants and model architecture used for training are consistent with those in Section VII-A. Experimental results show that PEFL reduces the computational cost by 20.86% compared to ShieldFL, while improving accuracy by 10.39%. Although the computational cost increases by 27.21% compared to ProFL, both accuracy and communication round are significantly improved. This higher cost is because PEFL employs TTHE combined with the blinding factor technique, which requires multiplicative homomorphic operations on two ciphertexts and is more computationally intensive than the linear homomorphic encryption used in ProFL. However, PEFL achieves an optimal balance between security and efficiency.

VII. SIMULATION EXPERIMENT AND ANALYSIS

In this section, we will evaluate the ability of PEFL framework to resist the backdoor and collusion attacks in MECS and verify the effectiveness of the proposed models and algorithms in improving the accuracy of global model of FL.

A. Experimental Setup

1) Datasets: PEFL was deployed in PyTorch and executed on a server equipped with 16-core CPUs, 24 GB RAM and an NVIDIA GeForce RTX 3080 Ti. The datasets were used to train and test the performances of PEFL.

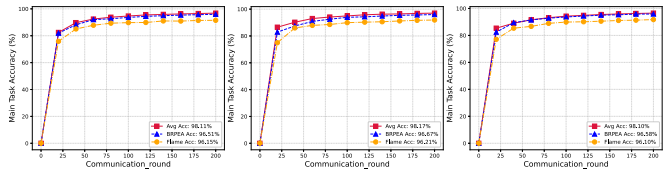
- *MNIST:* This dataset contains 70,000 samples of handwritten digits (0-9) and it is divided into 60,000 training images and 10,000 testing images.
- *CIFAR-10:* This dataset consists of 10 categories of RGB color images and it includes 50,000 training images and 10,000 test images.
- *FMNIST:* FMNIST can serve as an alternative to MNIST dataset and it contains many grayscale images of 10 different categories. The dataset includes 70,000 samples and every category consists of 6,000 samples in the training set and 1,000 samples in the test set.

2) Parameter and Evaluation Metrics: This experiment involved 100 participants, 15% of whom were malicious. In each FL training round, ESs randomly selected 20% of participants to perform 3 local iterations with a learning rate of 0.01. To ensure consistency with prior works [35] [37], a fixed random seed of 0 was used for all experiments. As described in Section V-A, we employed the DPEE method to add noise to local models. The noise distribution was defined by the privacy budget ϵ and the scoring function $q(\bar{\omega}_i^{\chi}(\kappa), \omega_i^{\chi}(\kappa))$ as discussed in (14). These parameters determined the probability of selecting any value from the candidate set $\mathbb{S}$ in (13). For all datasets, a Convolutional Neural Network (CNN) architecture was employed, comprising two convolutional layers with 5×5 kernels followed by two fully connected layers with 512 neurons each.

We evaluate the scheme performance based on five metrics as follows.

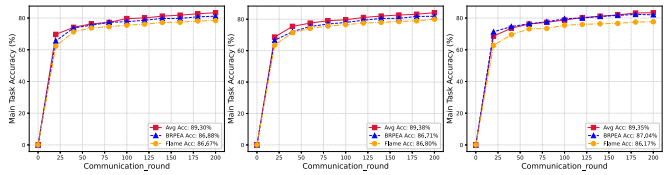
- *Poisoning Data Rate (PDR):* PDR is the proportion of data successfully poisoned by the attacker to total data.
- *Average Backdoor Success Rate (ABSR):* ABSR indicates the average of Backdoor Success Rate (BSR) for all iterations of global model.
- *Main-task Accuracy (MA):* MA is the accuracy of global model on the test dataset. The attacker hopes to reduce the impact on MA and avoid being identified by the defense mechanism.
- *Accuracy:* Adversarial test set accuracy of global model serves as a key metric to characterize the robustness of PEFL framework both collusion and backdoor attacks.
- *F1 score:* The F1 score considers both the precision and recall performance of the model, representing the harmonic mean of precision and recall and indicating the balance between precision and recall.

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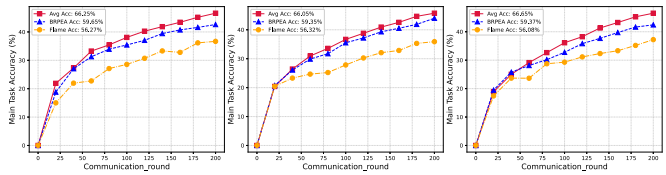
(a) PDR=15.63% (b) PDR=31.25% (c) PDR=46.88%

Fig. 4. Comparison of MA between different algorithms under different PDRs on MNIST.



(a) PDR=15.63% (b) PDR=31.25% (c) PDR=46.88%

Fig. 5. Comparison of MA between different algorithms under different PDRs on FMNIST.



(a) PDR=15.63% (b) PDR=31.25% (c) PDR=46.88%

Fig. 6. Comparison of MA between different algorithms under different PDRs on CIFAR10.

B. Results and Analysis

1) Ma and Absr of Brpea: We use the FedAvg [46] algorithm without any defense mechanism as a baseline and evaluate the defense performance of BRPEA and Flame [34] mechanisms when the Distributed Backdoor Attack (DBA) appears in MECS. DBA continuously injects backdoors to strength the concealment of attacks in the condition of maintaining the performance of MA. Therefore, the baseline FedAvg can still keep a high MA facing DBA.

Figs. 4 and 5 present the comparison of MA between FedAvg, Flame and BRPEA mechanisms on MNIST and FMNIST datasets under varied PDRs (15.63%, 31.25% and 46.88%). The results indicate that the performance difference between BRPEA and FedAvg in terms of MA is below 5% when the global model tends to converge. Noticed that BRPEA inevitably interferes some benign local models when it identifies some backdoor models, which results in a lower MA than the baseline algorithm FedAvg. BRPEA is superior to Flame in all PDRs. Taking PDR of being 15.63% on FMNIST as an instance, MA of BRPEA reaches 86.88% and that of Flame is 85.88%. This demonstrates that BRPEA achieves a better balance between the effectiveness of defense and the performance of MA.

MA of BRPEA is compared with the baseline FedAvg and Flame mechanisms under different PDRs on the complex dataset CIFAR10 in Fig. 6. The gap of MA between BRPEA and FedAvg is approximately 18% and it is nearly 5% on the simple datasets MNIST and FMNIST. This proves our BRPEA mechanism can work better in the high-dimensional complexity dataset and it may be applied in more practical scenarios. MA appears a decreasing trend in all mechanisms as PDR increases from 15.63% to 46.88%. This phenomenon further explains that backdoor attack on complex datasets can exponentially amplify the interference of defense mechanisms in normal learning process of FL global model.

TABLE V
ABSR COMPARISON OF DIFFERENT SCHEMES

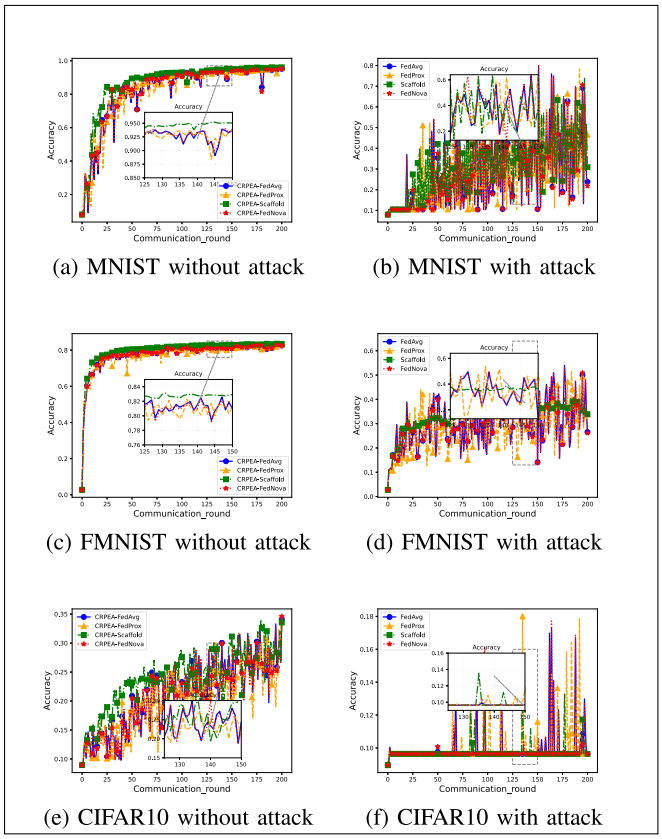
PDR	Mechanism	DP Type	Dataset		
			MNIST	FMNIST	CIFAR10
15.63%	FedAvg	—	100%	99.99%	90.10%
	FLAME	Gaussian	9.03%	12.48%	10.35%
	BRPEA	Exponential	0.21%	0.16%	7.77%
31.25%	FedAvg	—	100%	99.98%	91.43%
	FLAME	Gaussian	3.60%	9.37%	8.18%
	BRPEA	Exponential	0.40%	1.58%	8.48%
46.88%	FedAvg	—	100%	99.92%	93.15%
	FLAME	Gaussian	5.18%	11.62%	11.44%
	BRPEA	Exponential	0.19%	1.25%	7.15%

Table V compares the differences in ABSR of various defense mechanisms and also shows the privacy type adopted by each mechanism. ABSR for the undefended FedAvg is close to 100% across all datasets under different PDRs, indicating the global model is vulnerable to backdoor attacks. For Flame with the Gaussian mechanism on MNIST dataset, ABSR decreases from 9.03% to 3.60% as PDR increases from 15.63% to 31.25%. However, it rises to 5.18% when PDR is 46.88%, denoting performance fluctuations under different PDRs. In contrast, BRPEA with DPEE method demonstrates robust performance across all datasets. On the relatively simple datasets (MNIST and FMNIST), its ABSR remains below 1.58% and drops to 0.19% when PDR is 46.88%. On more complex CIFAR10 dataset, BRPEA maintains stable performance, with its ABSR ranging from 7.15% to 8.48%. Although a high PDR may increase ABSR, BRPEA exhibits a superior ability to distinguish between malicious and benign models.

In summary, compared with FedAvg and Flame, the proposed BRPEA can comprehensively improve the performances of MA and ABSR when defending against backdoor attack.

2) Accuracy and Loss of CRPEA: Fig. 7 compares the accuracy of four typical FL algorithms (FedAvg [46], FedProx [47], Scaffold [48] and FedNova [49]) against collusion attack on three datasets. In the defense scenarios under collusion attack, the differences in accuracy among all FL algorithms on MNIST and FMNIST datasets are no more than 5%. Scaffold shows a better performance due to its gradient correction mechanism. The accuracy of all algorithms decreases by approximately 25% on the complex CIFAR10 dataset, which denotes the data dimension and distribution have a markable impact on the model performance. The performance of each algorithm rapidly deteriorates if a defense mechanism is not adopted and the training process of FL exhibits undetermined fluctuations. What's worse, this may cause the failed convergence of global model due to the incorrect parameter updating.

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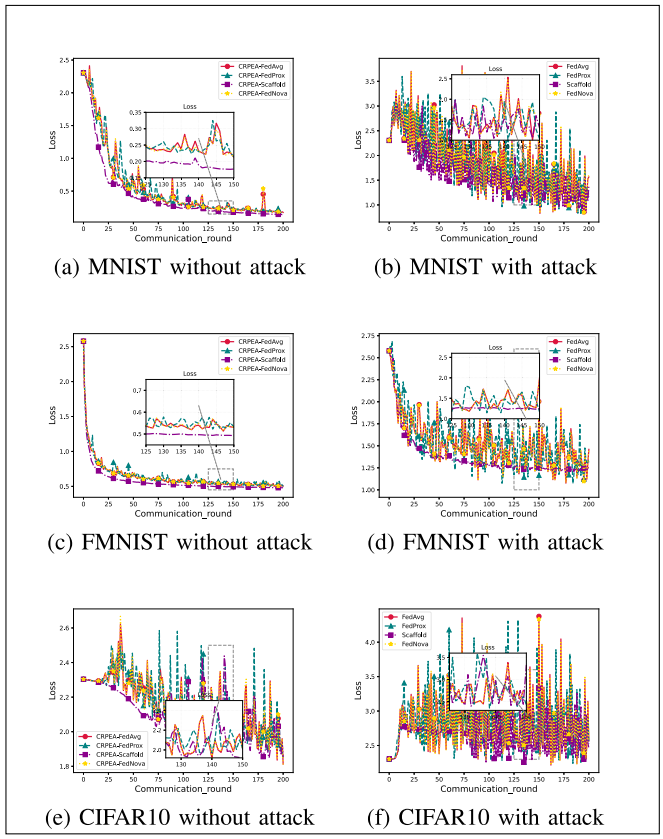


(a) MNIST without attack (b) MNIST with attack

(c) FMNIST without attack (d) FMNIST with attack

(e) CIFAR10 without attack (f) CIFAR10 with attack

Fig. 7. Accuracy comparison of four FL algorithms.



(a) MNIST without attack (b) MNIST with attack

(c) FMNIST without attack (d) FMNIST with attack

(e) CIFAR10 without attack (f) CIFAR10 with attack

Fig. 8. Loss comparison of four FL algorithms.

As illustrated in Fig. 7(b), collusion attack result in drastic deviation of accuracy in each algorithm on MNIST dataset. Scaffold displays a superior stability in accuracy (less than 10%). In contrast, the accuracy of FedNova changes by more than 43% and that of FedAvg sharply drops from 64.52% to 24.45%. Although FedProx introduces a proximal term to suppress the gradient deviation, the accuracy is still unstable and it rises from 35.18% to 52.67%. In experiments on FMNIST and CIFAR10 datasets in Fig. 7(d) and (f), the accuracy of Scaffold maintains at 35% and 9.8%, respectively. Compared with FedNova and FedAvg, the accuracy of Scaffold is relatively stable and the changing rate does not exceed 5%. Collusion attack seriously decreases the stability and the aggregation accuracy and our CRPEA mechanism with Scaffold algorithm can improve the stability, accuracy and robustness of FL models in MECS.

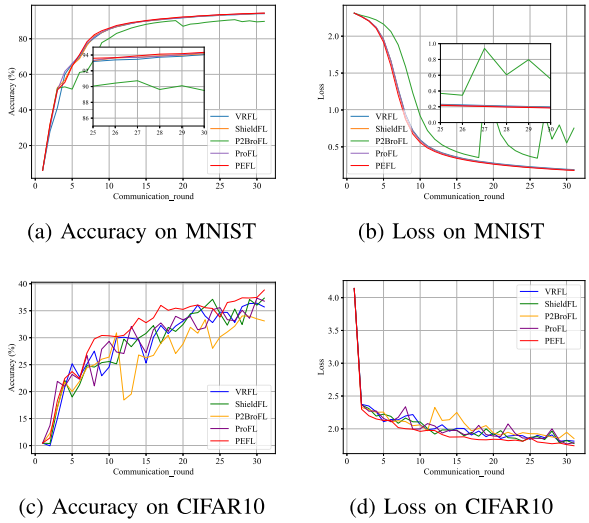
Fig. 8 illustrates the loss of CRPEA with four different FL algorithms (FedAvg, FedProx, Scaffold and FedNova) against collusion attack on three datasets. In the defense scenario with collusion attack, the loss difference between algorithms on MNIST and FMNIST does not exceed 0.05 and all algorithms converge in about 100 training rounds. The loss increases by 0.2 and the convergence of algorithms extends at the 180-th training rounds on the complex dataset CIFAR10.

As illustrated in Fig. 8(d), Scaffold shows a stable convergence through a gradient correction mechanism and effectively limits the change of loss within 0.02 on FMNIST. Conversely, FedNova, FedAvg and FedProx algorithms exhibit a greater deviation in loss due to the interference caused by collusion attack. Additionally, Scaffold demonstrates the most stable performance in loss on CIFAR10 as shown in Fig. 8(f). The loss of other algorithms are dramatically changed even exceeding 4.0, especially the loss of FedNova. The global model converges after the 200-th round for all algorithms. This verifies that the robust design of Scaffold in CRPEA can effectively resist collusion attack.

The experimental results in Figs. 7 and 8 prove that CRPEA can enhance the resistance of FL in MECS against collusion attack. All algorithms can converge in expected training rounds and demonstrate the preferred performance in terms of accuracy, loss value, security and robustness with the assistance of CRPEA defense mechanism.

3) Performance Comparison: We evaluate PEFL with four mechanisms, ShieldFL [35], VRFL [50], P2BroFL [36] and ProFL [37], in terms of accuracy and loss on MNIST and CIFAR10 datasets. As shown in Fig. 9, while ShieldFL, ProFL, VRFL and P2BroFL demonstrate comparable performance, our proposed PEFL outperforms them from the overall perspective of accuracy and loss on all datasets. This superior performance highlights the effectiveness of PEFL in defending against both backdoor and collusion attacks in MECS. It is worth noting that existing mechanisms typically address either backdoor or collusion attacks separately, whereas our proposed PEFL provides comprehensive protection against both attack types without losing accuracy, security and privacy of MECS.

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(a) Accuracy on MNIST (b) Loss on MNIST

(c) Accuracy on CIFAR10 (d) Loss on CIFAR10

Fig. 9. Accuracy and loss for different schemes.

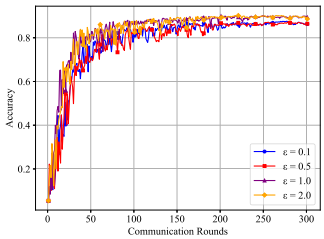
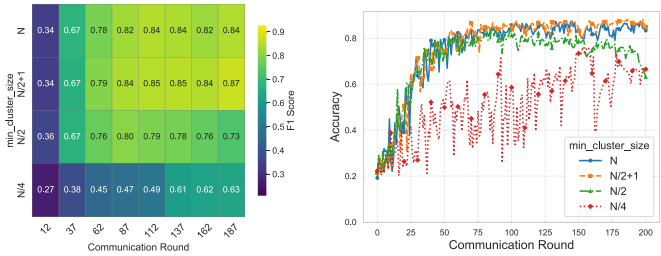


Fig. 10. Impact of different privacy budgets ϵ on accuracy.

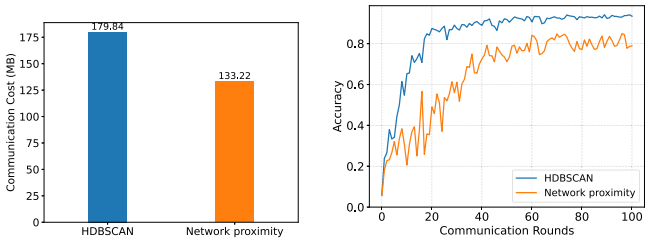


(a) F1 score heatmap (b) Accuracy comparison

Fig. 11. Ablation analysis of different minimum cluster sizes.

4) Ablation Experiment: We evaluate the impact of different ϵ (0.1-2.0) on model accuracy and analyze its sensitivity with respect to communication rounds T (100, 200 and 300), as shown in Fig. 10. The results show that a smaller ϵ introduces excessive DPEE noise, which results in a decrease in accuracy. On the contrary, a larger ϵ improves the accuracy but weakens the privacy preserving. Furthermore, the model does not fully converge with fewer communication rounds especially when $T < 50$, and more communication rounds can promote the stability of model performance. A better balance between accuracy and privacy preserving is obtained when $\epsilon = 0.5$ and $T \geq 200$.

Fig. 11 presents the ablation analysis on the HDBSCAN parameter, $min_cluster_size$ (minimum cluster size). Fig. 11(a) compares the changing in F1 Score across communication rounds for different minimum cluster sizes ($N/4$, $N/2$, $N/2+1$ and N). The results that a high F1 score is maintained in most of communication rounds when $min_cluster_size \geq N/2+1$. Fig. 11(b) further verifies that HDBSCAN exhibits more stable accuracy under the same condition.



(a) Communication cost (b) Accuracy

Fig. 12. Comparison of communication cost and accuracy of different clustering strategies.

We evaluate the impact of clustering strategies on the performance of PEFL by comparing approaches based on model similarity and network proximity. As shown in Fig. 12, HDBSCAN incurs a higher communication cost than network proximity clustering (179.84 MB vs. 133.22 MB), but it achieves higher model accuracy. Although grouping geographically distant clients by model similarity increases communication costs, HDBSCAN maintains more robust performance.

VIII. Conclusion

This paper proposed a privacy-enhanced federated learning framework when collusion and backdoor attacks were launched in an MECS system. BRPEA defense mechanism based on DPEE and HDBSCAN clustering method with adaptive clipping technique was proposed in PEFL which effectively relieved the negative effects of external attacks on participants, prevented the global model from being misled and improved the accuracy of MECS. In addition, CRPEA mechanism in PEFL employed TTHE combined with HKE protocol to generate the blinding factors and protect the privacy of local models of FL. Experimental results prove that PEFL significantly promotes the robustness against both backdoor and collusion attacks while ensuring the accuracy of aggregated global model of FL. Future work will focus on improving the capability of PEFL against diverse attacks and optimizing the communication efficiency of FL in MECS.

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